# Joseph Wardle - Production TD

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## Work Experience

**Pipeline Technical Director - Student Short Films** October 2025 - Present
Brigham Young University Center for Animation *Provo, UT*

- Develop and support **Python/Qt** tools and a **USD** pipeline for **50+ artists** on Linux and Windows; troubleshoot DCC tools and publishing failures, and manage shot and asset data.

**Sandwich Kwon Do**

- Implemented **immutable, versioned USD publishes** and pinned every layer's version at **Tractor** submission, eliminating flicker caused by assets republished mid-render.
- Built a **Maya previs-to-layout workflow** with dry-run validation, **ShotGrid** shot creation and re-cutting, and versioned layout scene saves.
- Built **one-click Maya publishing** through a headless Houdini USD asset builder shared with Substance Painter.
- Built a shared **Maya/Houdini playblast previewer** with FFmpeg burn-ins and **ShotGrid Version uploads**.
- Established the show's **OCIO** color pipeline, baking **OpenDRT and ACES 2.0** to $65^3$ LUTs for OCIO 2.3 compatibility across Houdini, Maya, Nuke, and Blender.

**Honey Business**

- Built whole-collection **VDB-to-mesh foliage proxies**, switched with USD purposes, for **up to 70× faster viewport performance**.

**Web Developer** March 2025 - November 2025
Brigham Young University - Computer Science Department *Provo, UT*

- Developed and maintained **Django** web services, REST endpoints, and internal admin UIs for non-technical users.
- Built **GitHub Actions** pipelines for testing, container builds, and deployment; used **Docker Compose** for local development and **Jira** for project tracking.

## Projects

**Cenote - GPU Path Tracer and Hydra Render Delegate** github.com/joseph-wardle/cenote

- Built a **Rust/Vulkan** GPU path tracer with **Slang** kernels and a **Hydra 2** render delegate for usdview, husk, and Houdini.
- Implemented **OpenPBR** shading with baked energy-compensation tables; verified with white furnace tests and comparisons against **pbrt-v4**.

**Javelin - Rigid Body Physics Engine** (github.com/joseph-wardle/javelin) 2023 - Present

- Built a **C++23** simulator using named modules, fixed 60 Hz physics on a dedicated thread, BVH broad phase, and a warm-started constraint solver; added an **ImGui/OpenGL** viewer and **Tracy** profiling.

**Film Grain Synthesis** (github.com/joseph-wardle/film_grain) August 2025

- Reimplemented Newson et al. (2017) in **Rust** with **Rayon** CPU parallelism and **wgpu** compute; multithreaded CPU is a **median 72× faster** than the reference C implementation (~7,200 configurations, ~56,000 renders; median of three runs each).

## Education

**Brigham Young University** Provo, UT
*Bachelor of Science, Computer Science (Animation & Games Emphasis)* *August 2023 - Expected May 2027*

- **GPA: 3.95.** Relevant coursework: Data Structures, Multithreading, Linear Algebra, Modeling, Rigging, Shading, FX.

## Skills

- **Programming:** Python, C++23, Rust, MEL, TypeScript, Bash
- **Pipeline:** USD, ShotGrid, Pixar Tractor, OCIO, RenderMan, MoonRay
- **DCCs & UI:** Maya, Houdini, Nuke, Substance Painter, Blender, Qt/PySide2, ImGui
- **Development:** Git, Perforce, GitHub Actions, Jira, Django, Docker; Linux, Windows